



GTFKAN: A Novel Microbe-drug Association Prediction Model Based on Graph Transformer and Fourier Kolmogorov-Arnold Networks

Jiacheng Lai, Zhen Zhang, Bin Zeng, and Lei Wang*

Technology Innovation Center of Changsha, Changsha University, Changsha 410022, China

Big Data Innovation and Entrepreneurship Education Center of Hunan Province, Changsha University, Changsha 410022, China

Correspondence to Lei Wang: Technology Innovation Center of Changsha, Changsha University, Changsha 410022, China. wanglei@xtu.edu.cn (L. Wang)

<https://doi.org/10.1016/j.jmb.2025.169201>

Edited by Rita Casadio

Abstract

Microbes have been shown to be closely related to human health. In recent years, lots of computational methods for predicting microbial-drug association have been proposed. In this manuscript, we introduced a novel predictive model, called GTFKAN, to identify potential microbe-drug associations by combining Graph Transformation Networks (GTN) with Fourier Kolmogorov-Arnold Networks (FKAN). In GTFKAN, we would first compute the Gaussian kernel and functional similarity of microbes and drugs respectively, and then adopt random walk and restart (RWR) methods to enhance these similar features to construct a new microbe-drug heterogeneous network HN. At the same time, we would further calculate the cosine similarity of microbes and diseases to construct another microbe-drug heterogeneous network LDIM. Next, we would input HN into GTN to derive the location and structural features of microorganisms and drugs, and input LDIM into FKAN to extract the hidden higher-order features of microorganisms and drugs, respectively. Finally, we would integrate these two features extracted by GTN and FKAN and feed the integrated features into the MLP classifier to infer potential microbial-drug associations. Moreover, to evaluate the performance of GTFKAN, we compared it with state-of-the-art methods based on well-known public datasets, and the experimental results show that GTFKAN can achieve satisfactory predictive performance. In addition, the results of ablation experiments and case studies also demonstrated the superiority of GTFKAN, which means that GTFKAN may be a useful microbial-drug association prediction tool in the future.

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Introduction

More and more evidences have demonstrated that microbes can influence the metabolism and efficacy of drugs and alter clinical outcomes through their metabolic activity [1]. For instance, *Eggerthella lenta* can metabolize the cardiac drug digoxin, which reduces its effective concentration and therapeutic efficacy in heart failure patients [1]. Intestinal bacteria, including *Enterobacter* and

Bifidobacterium, affect the metabolism of the anti-cancer drug irinotecan, impacting its efficacy and side effect profile [2,3]. Furthermore, the impact of microorganisms is notably pronounced in antibiotic use [4]. Vancomycin, effective against certain infections, can disrupt the normal gut microbiota and heighten the risk of *Clostridioides difficile* infection [5]. Research indicates a marked rise in *Clostridioides difficile* incidence among patients on broad-spectrum antibiotics, prompting greater caution in

clinical prescribing practices [6–8]. In psychiatric disorders, research reveals a robust link between gut microbiota composition and the potency of antidepressants like fluoxetine [9,10]. Specific microbes, including *Lactobacillus*, potentially amplify the efficacy of antidepressants through modulation of neurotransmitter synthesis [11].

Microbial-drug interactions, while rich in research potential, are often limited by traditional wet-laboratory methods due to high cost, time-consuming, and difficult microbial cultures [12]. Laboratory studies are often time-consuming and resource-intensive in assessing drug effects on microorganisms [13]. These limitations underscore the urgent need for innovative approaches to detect and analyze the intricate relationships between microbes and drugs [14]. As bioinformatics and artificial intelligence advance, computational predictive models demonstrate significant potential for identifying microbe-drug associations [15,16]. These models efficiently process large biomedical datasets, prioritize experimental design, and advance drug development and disease diagnosis [17,18]. This study explores the significance of microorganisms in medicine and drug development, the intricacies of microbial-drug interactions, and the potential of computational methods, offering innovative directions for future research.

Experimental data were sourced from multiple public microbial-drug association databases. MDAD [19], aBiofilm [20], and DrugVirus [21] house extensive, experimentally validated microbial-drug association data, forming a crucial basis for developing and validating machine learning models. MDAD records microbial-drug associations, aBiofilm includes antimicrobial biofilm characterization data, and DrugVirus concentrates on drug-virus interactions. Utilizing these resources enhanced experimental reliability and substantially decreased the cost and time associated with traditional wet lab experiments. For instance, Fan et al. developed the MDASAE computational method, utilizing a stacked autoencoder and multi-head attention mechanism to predict potential microbial-drug associations [22]. Tan et al. utilized singular value decomposition and Node2vec for analyzing microbial-drug associations [23]. Yang et al. employed Learning Graph-based Convolutional Attention Networks and Self-paced Iterative Sampling (LCASPMDA) for predicting microbial-drug associations [24]. Kuang et al. integrated global and local features into the GLNNMDA computational model for predicting potential microbial-drug associations [25]. Zhang et al. introduced the GCANCAE model, enhancing the feature representation of GCN and CSAE on heterogeneous networks to predict potential microbial-disease associations, by constructing comprehensive feature representations and calculating the inner product of matrices for disease and microorganism features [26]. Researchers have not only achieved outstanding results in

microbe-drug association prediction but also offer valuable methods for related fields, including microbe-disease and circRNA-disease association predictions [27]. Lan et al. proposed a Knowledge Graph Attention Network (KGANCA) to predict circRNA-disease associations [28]. Wu et al. also conducted similar work, embedding data with a transformer encoder. Subsequently, an improved knowledge graph (KGETCDA) was employed to predict circRNA-disease associations [29]. Likewise, Li et al. introduced a model (DWNCPDA) predicated on deep walk and network consistency projection for predicting circRNA-disease associations [30]. Li et al. extracted key topological and functional protein features from weighted PPI networks and homologous protein data (CVIM) to identify essential proteins in brewer's yeast [31]. Zhu et al. introduced a protein evolutionary whole process network (MWPNE) for identifying essential proteins [32].

Different from existing models, we introduced a new hybrid neural network, GTFKAN, which utilizes graph converters and Fourier-Kolmogorov-Arnold networks to infer potential microbial-drug interactions. In GTFKAN, we first calculated the Gaussian kernel and functional similarity of drugs and microorganisms separately to construct heterogeneous microbe-drug networks, and then calculated cosine similarity to form two low-order feature matrices of drugs and microorganisms respectively. Next, we fed the heterogeneous microbe-drug network into GTN, and extracted the local and global node location information and edge structure characteristics of each microbe and drug node in the network. And at the same time, we further inputted those two low-order feature matrices into FKAN to learn their higher-order features separately. Thereafter, we integrated these two kinds of features extracted by GTN and FKAN respectively to form a comprehensive feature matrix, based on which, we finally utilized the MLP classifier to predict potential microbial-drug associations.

Materials and Methods

Datasets

In this section, we would first download known Microbe-Drug associations from two well-known public databases, such as the MDAD and the aBiofilm databases, respectively. Among them, the MDAD database contains 1388 drugs and 180 microorganisms, and the aBiofilm database consists of 1720 drugs and 140 microorganisms. After eliminating duplicates, we finally obtained 2470 known Microbe-Drug associations between 1373 drugs and 173 microorganisms from MDAD (<https://chengroup.cumt.edu.cn/MDAD>), and 2884 known Microbe-Drug associations from aBiofilm (<http://bioinfo.imtech.res.in/manojk/abiofilm/>),

respectively. Next, as we needed to calculate the functional similarity of microbes and drugs, based on these 1373 drugs and 173 microorganisms downloaded from MDAD, we further manually compiled 1122 known Drug-Disease associations between these 1373 drugs and 109 diseases, and 402 known Microbe-Disease associations between these 173 microbes and 109 diseases based on the method proposed by Wang et al. [33]. And at the same time, based on these 1720 drugs and 140 microorganisms downloaded from aBiofilm, we as well artificially obtained 6749 known Drug-Disease associations between 1720 drugs and 72 diseases and 722 known Microbe-Disease associations between 140 microbes and 72 diseases. Table 1 provided a detailed breakdown of the data downloaded from these two databases.

Method overview

As shown in Figure 1, GTFKAN mainly includes the following four steps:

Step1: Constructing two heterogeneous networks HN and LDIM based on multiple similar indicators of microorganisms and diseases respectively.

Step2: Adopting GTN to extract node location information and edge structure features of microbial and drug sub-graphs in HN.

Step3: Adopting FKAN to capture the high-order features of microorganisms and drugs in LDIM.

Step4: Integrating the above two kinds of features of microbes and drugs, and feeding the integrated features into the MLP classifier to obtain the predicted scores of potential microbe-drug associations.

Construction of the adjacency matrix A . In this section, we would first construct an adjacency matrix $A \in R^{n_r \times n_m}$ based on above newly-downloaded known microbe-drug associations as follows: for any given drug r_i and microbe m_j , there is $A_{ij} = 1$ if and only if there is a known association between r_i and m_j , otherwise there is $A_{ij} = 0$. Here, n_r and n_m represent the number of drugs and microbes, respectively. Subsequently, we would calculate various similarities of microbes and drugs separately. Finally, we would integrate these newly constructed microbial and drug similarity networks, adjacency matrices to construct microbial-drug

heterogeneous networks and low-dimensional feature matrices simultaneously.

Calculation of the Gaussian kernel similarity of microbes and drugs. Let $L(i)$ and $L(j)$ represent the i -th and j -th rows of the adjacency matrix A respectively, the Gaussian kernel similarity between any two drugs r_i and r_j can be computed according to the following formulas (1) and (2):

$$GIR(r_i, r_j) = \exp(-\mu \|L(i) - L(j)\|^2) \quad (1)$$

$$\mu = \frac{1}{\frac{1}{n_r} \sum_{i=1}^{n_r} \|L(i)\|^2} \quad (2)$$

Similarly, let $M(i)$ and $M(j)$ denote the i -th and j -th columns of the adjacency matrix A , respectively. For any two microbes m_i and m_j , the Gaussian kernel similarity between them can be estimated according to the following formulas (3) and (4):

$$GIM(m_i, m_j) = \exp(-\gamma \|M(i) - M(j)\|^2) \quad (3)$$

$$\gamma = \frac{1}{\frac{1}{n_m} \sum_{i=1}^{n_m} \|M(i)\|^2} \quad (4)$$

Calculation of the functional similarities between microbes or drugs. In this section, we would adopt those newly-downloaded drug-disease, microbe-disease, and microbe-drug associations to calculate the functional similarity between microbes and drugs, based on the method proposed by Zhou et al. [34] in the following way: First, we would extract semantic disease information from the disease ontology [35], which compiles common human diseases based on classification standards like MeSH (<https://www.ncbi.nlm.nih.gov/mesh/>) and ICD (<https://icd.who.int/browse11/l-m/en>), and the disease information is represented by directed acyclic graph topology to illustrate hierarchical relationships. And then, we would calculate the semantic similarity of diseases based on the method proposed by Wang et al. [36] as follows: for any given disease d_i , let D_i be the set including d_i and its related diseases, then the semantic contribution values of all subitems in D_i can be obtained according to the following formula (5):

$$H_{d_i}(d) = \begin{cases} 1, & d = d_i \\ \max_{d' \in \text{childrenof}} \left(\frac{F_{d_i}(d')}{2} \right), & d \in D_i \text{ and } d \neq d_i \end{cases} \quad (5)$$

Table 1 The statistics of the MDAD and the aBiofilm databases.

Datasets (MDAD/aBiofilm/Sum of the two)	Microbes	Diseases	Drugs	Associations
Microbe-Drug associations	173/140/199	\	1373/1720/1953	2470/2884/3476
Microbe-Disease associations	173/140/199	109/72/114	\	402/ 722/874
Drug-Disease associations	\	109/72//114	1373/1720/1953	1122/ 6749/7347

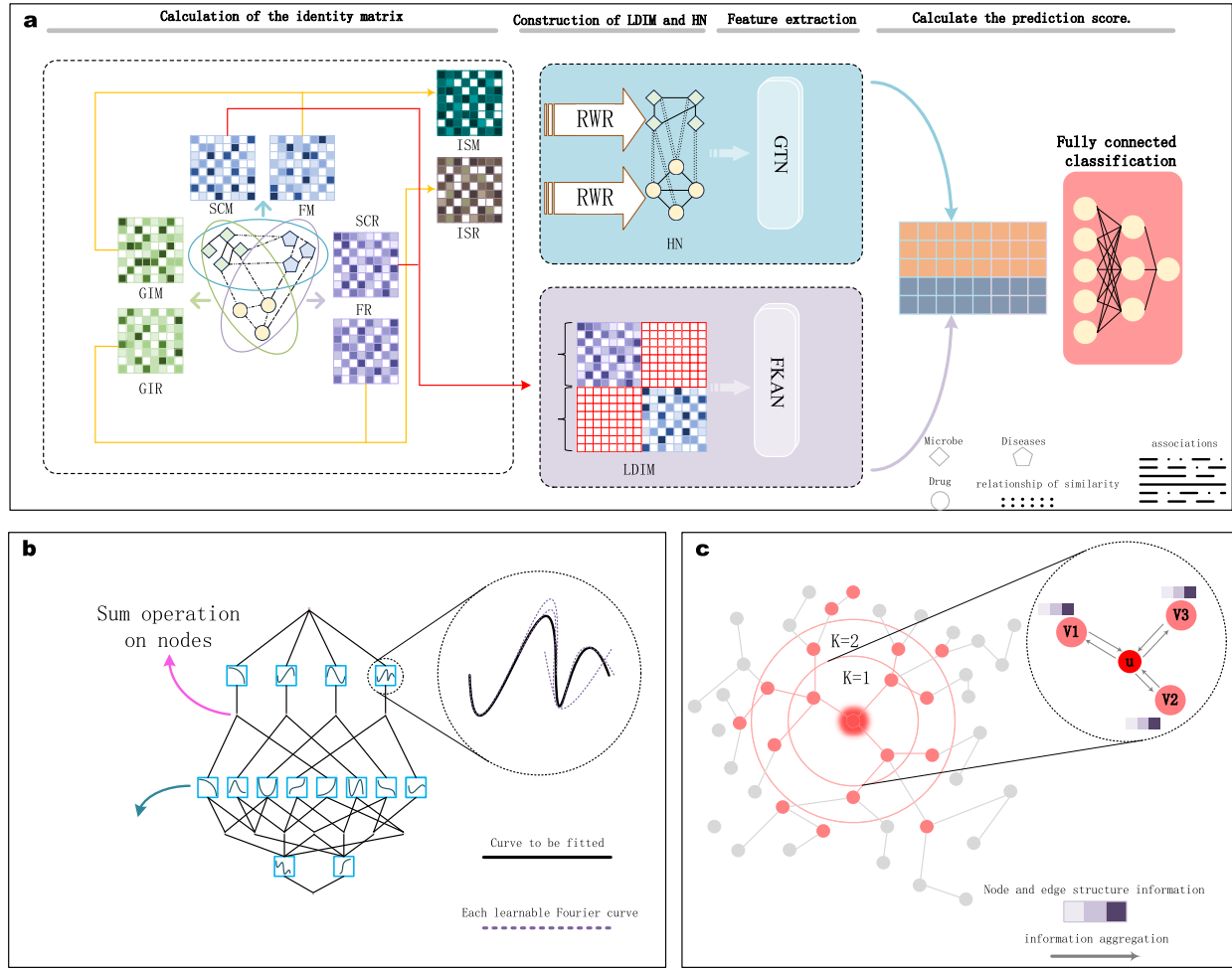


Figure 1. Overview of the GTFKN Framework. (a) Constructing two heterogeneous microbe-drug networks HN and LDIM, based on which, extracting features of microbes and drugs by adopting GTN and FKAN, respectively. (b) The feature extraction process of FKAN. The activation functions are positioned on the edges and fitted by using the Fourier series. (c) The feature extraction process of GTN. Construction of the microbe-drug heterogeneous network and low-order features matrix.

Next, for any given disease d_j other than d_i , the semantic similarity between them can be calculated according to the following formula (6):

$$DS(d_i, d_j) = \frac{\sum_{d \in D_1 \cap D_2} (H_{d_i}(d) + H_{d_j}(d))}{DM(d_i) + DM(d_j)} \quad (6)$$

In above Eq. (6), $DM(d_i) = \sum_{d \in D_1} F_{d_i}(d)$ represents the sum of all subitem semantic contribution values for D_1 .

Obviously, the disease semantic similarity defined above described the overlap of related diseases between two diseases, indicating that the more closely related diseases have greater weight.

Moreover, for any two given drugs r_i and r_j , let d_{i1} and d_{j2} represent the number of diseases associated with r_i and r_j , respectively, then he

drug functional similarity $FR(r_i, r_j) \in R^{n_r \times n_r}$ between them can be calculated according to the following formula (7):

$$FR(r_i, r_j) = \frac{1}{b_1 + b_2} \left[\sum_{t=1}^{b_1} \max_{1 \leq s \leq b_2} (DE(d_{i1}, d_{j2})) + \sum_{s=1}^{b_2} \max_{1 \leq t \leq b_1} (DE(d_{i1}, d_{j2})) \right] \quad (7)$$

Similarly, for any two given microorganisms m_i and m_j , let d_{i3} and d_{j4} denote the number of diseases associated with m_i and m_j , respectively, then the microbial functional similarity $FM(r_i, r_j) \in R^{n_m \times n_m}$ between them can be calculated according to the following formula (8):

$$FM(m_i, m_j) = \frac{1}{n_1 + n_2} \left[\sum_{p=1}^{n_1} \max_{1 \leq k \leq n_2} (DE(d_{i3}, d_{j4})) + \sum_{k=1}^{n_2} \max_{1 \leq p \leq n_1} (DE(d_{j4}, d_{i3})) \right] \quad (8)$$

Calculation of the integrated similarities between microbes or drugs. Thereafter, by integrating above newly-obtained microbial and drug similarities, we can derive a drug similarity synthesis matrix (ISR) and the microbial similarity synthesis matrix (ISM) according to the following formulas (9) and (10) respectively:

$$ISR(r_i, r_j) = \begin{cases} \frac{GIR(r_i, r_j) + FR(r_i, r_j)}{2}, & \text{if } FR(r_i, r_j) \neq 0 \\ GIR(r_i, r_j), & \text{otherwise} \end{cases} \quad (9)$$

$$ISM(m_i, m_j) = \begin{cases} \frac{GIM(m_i, m_j) + FM(m_i, m_j)}{2}, & \text{if } FM(m_i, m_j) \neq 0 \\ GIM(m_i, m_j), & \text{otherwise} \end{cases} \quad (10)$$

Furthermore, to effectively capture the similarity and structural information between microbial and drug nodes for identifying similar sub-graphs, we would further apply the Random Walk with Restart (RWR) algorithm to the ISR and ISM matrices respectively according to the following formulas (11) and (12).

$$q_i^{l+1} = \lambda G q_i^l + (1 - \lambda) \xi_i \quad (11)$$

$$\xi_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

In above equations, ξ represents the restart probability, G denotes the transition probability matrix, $\xi_i \in R^{(1 \times m)}$ is the initial probability vector for node i , and q_i^l is the probability of node i transitioning to node l .

Obviously, based on the RWR process described above, we can derive a new drug similarity matrix S_r and a new microbe similarity matrix S_m based on the ISR and ISM matrices separately.

Hence, by combining these two matrices S_r and S_m with the adjacency matrix A , we can construct a novel heterogeneous microbe-drug network $HN \in R^{(n_r+n_m) \times (n_r+n_m)}$ according to the following formula (13):

$$HN = \begin{bmatrix} S_r & A \\ A^T & S_m \end{bmatrix} \quad (13)$$

where A^T represents the transpose matrix of A .

Calculation of the cosine similarity of microorganism and drug. In this section, we would further derive an adjacency matrix $D_r \in R^{n_r \times n_r}$ based on the newly-acquired known drug-drug associations, where n_r represents the number of newly downloaded different drugs. And then, for any two given drugs i and j in D_r , we would calculate the cosine similarity $SCR(i, j)$ between them according to the following formula (14):

$$SCR = \cos(D_r(i), D_r(j)) = \frac{D_r(i) \cdot D_r(j)}{\|D_r(i)\| \times \|D_r(j)\|} \quad (14)$$

Moreover, let n_r denote the number of newly downloaded different microbes, we would derive an adjacency matrix $D_m \in R^{n_m \times n_m}$ based on the

newly-acquired known microbe-microbe associations, and then, for any two given microbes i and j in D_m , we could also obtain the cosine similarity $SCM(i, j)$ between them in a similar way to above Eq. (14).

Thereafter, by integrating the two matrices SCR and SCM , we can construct a heterogeneous microbe-drug network $LDIM \in R^{(n_r+n_m) \times (n_r+n_m)}$ according to the following formula (15):

$$LDIM = \begin{bmatrix} SCR & 0 \\ 0 & SCM \end{bmatrix} \quad (15)$$

Feature extraction of nodes in HN with GTN

Current GNN interpretation methods concentrate on identifying key nodes, edges, and features, while ignoring sub-graph structures. Indeed, sub-graph structures offer a more intuitive understanding, with subgraph-level representations highlighting nearest neighbors, structure, and position, thus capturing both local and global graph information. Recently, the Transformer model is becoming increasingly popular in graph representation learning [37–39], for instance, Chen et al. introduced a Graph Transformer that incorporates a novel attention mechanism leveraging subgraph representations at each node, integrating structural information into the Transformer framework [38]. GTN merges the benefits of graph neural networks with the Transformer, effectively capturing node location and edge structure features. Consequently, this study employs GTN to learn the structural information within microbial-drug heterogeneous graphs, including inter-node positional information (The overall flow of GTN is shown in Figure 2).

Self-attention as kernel smoothing. In the Transformer model, input node features X are projected into query (Q), key (K), and value (V) matrices via linear transformations, where $Q = XW_Q$, $K = XW_K$ and $V = XW_V$, respectively. And besides, the transformer's attention mechanism, reformulated as a kernel smoother [40], employs a trainable exponential kernel based on node features to capture the attribute similarities between node pairs, which can be defined as the following formula (16):

$$Attn(x_v) = \sum_{u \in V} \frac{\mathcal{K}_{graph}(S_G(v), S_G(u))}{\sum_{u' \in V} \mathcal{K}_{graph}(S_G(v), S_G(u'))} f(x_u), \forall v \in V \quad (16)$$

where $S_G(v)$ denotes the subgraph in G centered on node v associated with node feature X in graph G . where $f(x_u) = W_V x$ represents the linear value function, with W_V being a trainable parameter. and $\mathcal{K}_{graph}$ can be any kernel for comparing pairs of subgraphs. This new self-attention function considers both node positional similarity and structural similarity between subgraphs, which can be defined as the following formula (17):

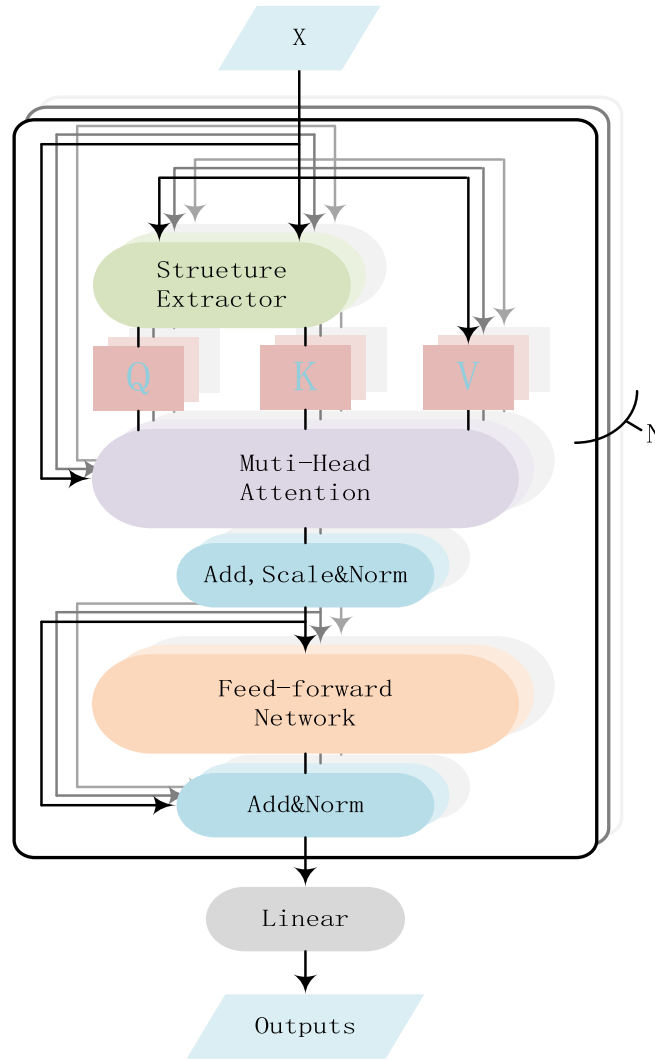


Figure 2. Flowchart of Graph Transformer.

$$\mathcal{K}_{\text{graph}}(S_G(v), S_G(u)) = \mathcal{K}_{\text{exp}}(\varphi(v, G), \varphi(u, G)) \quad (17)$$

where $\mathcal{K}_{\text{exp}}$ is a (non – symmetric) exponential kernel, which can be expressed in terms of the W_Q and W_K parameters as the following formula (18):

$$\mathcal{K}_{\text{exp}} = \exp\left(\frac{W_Q X, W_K X'}{\sqrt{d_{\text{out}}}}\right) \quad (18)$$

Besides, $\varphi(v, G)$ is a structure extractor that extracts a vector representation of some u centered subgraph with node features X , which can be defined as the following formula (19):

$$\varphi(v, G) = GCN^{(k)}(v) \quad (19)$$

where k denotes the number of layers in the model.

Encoding of graph structures in self-attentive mechanisms. Generally, the Transformer encoder is composed of N stacked layers, and each stacked layer includes a self-attention layer and a feed-forward network (FFN) layer. Besides, a

return layer will be presented before and after each FFN, utilizing ReLU as the activation function to prevent overfitting and enhance the model's generalization, and residual connections will be employed between these two sub-layers. The detailed computational process of the Transformer encoder can be defined as the following formulas (20) and (21):

$$X' = \text{LayerNorm}(X + \text{Sublayer}(X)) \quad (20)$$

$$X'' = \text{FFN}(X') = \text{ReLU}(X' W_1) W_2 \quad (21)$$

where X denotes the input node features, W_1 and W_2 are two trainable parameters.

Reconfiguration of heterogeneous networks. Firstly, given the ample feature information from subtrees learned via Graph Transformers, we would select a fully connected layer to aggregate these features and reconstruct the heterogeneous network. The output of the fully

connected layer can be defined as the following formula (22):

$$F = \text{ReLu}(X''W) + b \quad (22)$$

where W is a trainable parameter and b denotes the bias.

Additionally, we would employ the ReLu activation function to enhance the model's nonlinear representation and sparse activation. This enables the model to more effectively capture complex data structures, improving generalization and reducing the risk of overfitting [41]. Regarding optimizer selection, the model utilizes the Adam optimizer. Compared to the Adam optimizer, which integrates adaptive learning rates, momentum estimation, and effective regularization, addressing the regularization deficiencies of standard Adam and markedly enhancing the model's generalization performance [42]. It excels in deep learning tasks, particularly in training and fine-tuning large-scale models.

Capturing high-order features representations in LDIM using FKAN

In contemporary machine learning, feature extraction is essential during model training, particularly for processing low-order data [43]. Low-order data frequently harbor rich, latent information (high-order features), and effective feature extraction from these data can significantly impact the model's predictive power [44]. Among traditional feature extraction methods, the multilayer perceptron (MLP), a canonical deep learning model, has seen extensive use across various domains. However, with increasing data dimensions and complexity, MLP begins to exhibit limitations in managing complex features, particularly in the representation and prediction of high-dimensional data [45], where issues like overfitting and underrepresentation of features may arise [46].

Consequently, Kolmogorov-Arnold Networks (KANs), an emerging neural network architecture, overcome the limitations of MLPs and exhibit enhanced feature extraction capabilities due to their distinctive design [47]. KANs, drawing inspiration from the Kolmogorov-Arnold representation theorem [48], replace conventional linear weights with spline-parameterized univariate functions, allowing dynamic learning of activation patterns [49]. This adaptability enhances KANs' effectiveness in capturing complex data patterns and variations. In KANs, activation functions are applied to the edges, implying a structured transition of data as it traverses from neuron to neuron. This clarity allows us to observe how data traverses the network and the impact of each activation function on the data. This facilitates understanding of how the network processes input data, thereby enhancing model interpretability. KANs demonstrate theoretically and empirically superior neural scaling laws to MLPs, indicating greater efficacy in performance enhancement with increasing parameter counts.

Avoiding catastrophic forgetting: KANs leverage local plasticity to mitigate catastrophic forgetting, preventing the loss of previously learned knowledge when encountering new tasks, thus enhancing model generalization [50]. The Fourier transform enables efficient signal analysis in the frequency domain for KANs, further bolstering their feature extraction capabilities [51]. Via the Fourier transform, KANs can identify key frequency domain components, extract more representative features from high-dimensional data, and enhance predictive capabilities. Fourier-KAN (FKAN), a modification of the spline-based KAN employing Fourier series, results in more bounded values, thereby avoiding common network issues. Therefore, Fourier Kolmogorov-Arnold Networks are increasingly used in feature extraction [52].

A single Fourier-KAN (FKAN) layer is defined by the FKAN formula, applicable to arbitrary depths and widths, which can be defined as the following formula (23):

$$FKAN(H) = f(X) = \sum_{j=1}^{2(n_r+n_m)+1} \Phi_j \left(\sum_{i=1}^{n_r+n_m} \phi_{ij}(h_i) \right) \quad (23)$$

where ϕ_{ij} is a univariate continuous function mapping the input vector x , and Φ_j represents learnable activation functions. The FKAN implementation adheres to the Fourier series representation of the learnable activation functions, which can be defined as the following formula (24):

$$\phi_b(x) = \sum_{k=0}^G (a_k \cos(kx) + b_k \sin(kx)) \quad (24)$$

where a_k and b_k are the trainable Fourier coefficients, and G represents the grid size.

Fourier Kolmogorov-Arnold Networks (FKANs) comprise an encoder and a decoder, which can be defined as the following formulas (25) and (26):

$$h = FKAN_E(X) \quad (25)$$

$$Y = FKAN_D(h) \quad (26)$$

Let X represent the input low-order features and Y the output high-order features (HOF).

In FAKN, we employed the MSE loss function and the Adam optimizer for model optimization, which is defined as the following formula (27)

$$L_{\text{sparse}} = \frac{1}{n_r + n_m} \sum_{k=1}^{n_r+n_m} \|HOF(k) - LDIM(k)\|^2 + \beta P_{\text{penalty}} \quad (27)$$

Here, β is the weight of the sparsity penalty, set to 0.1. $HOF(k)$ and $LDIM(k)$ represent the k -th rows of HOF and $LDIM$, respectively.

Obtaining predicted scores by the multi-layer perceptron neural network

We can frame microbe-drug association prediction as a binary classification task, for which

we will design a multilayer perceptron (MLP) neural network classifier (The categorization process is shown in Figure 3.) to calculate the potential microbe-drug association scores. The MLP classifier will utilize a microbe-drug feature matrix as input. MLP contains various hidden layers that transmit feature weights from one layer to the next, which can be defined as the following formula (28):

$$M_l = \text{ReLU}(M_{l-1}W + b) \quad (28)$$

In the aforementioned equation, b represents the bias, W denotes the weights between the input and hidden layers, and the hidden layers are indexed as $l = 1, 2, \dots, k$. we incorporated three hidden layers into the model for classification, ultimately yielding the microbe-drug association probability $\hat{y}_{score}$. We employed cross-entropy as the loss function and the Adam optimizer for MLP optimization. The cross-entropy loss function is defined as the following formula (29):

$$\text{loss} = - \sum y_{\text{vectors}} \log \hat{y}_{score} + (1 - y_{\text{vectors}}) \log (1 - \hat{y}_{score}) \quad (29)$$

where y_{vectors} denotes a vector of the adjacency matrix A after being flattened.

Results

In this section, we would assess the predictive performance of GTFKAN under the framework of five-fold cross-validation and metrics including AUC, AUPR, accuracy, and recall. Initially, we would fine-tune the model parameters of GTFKAN to optimize performance across various microbe-drug datasets. Subsequently, we would conduct ablation studies to determine the impact of each model component on the predictive performance of GTFKAN. Next, we would compare of GTFKAN with four state-of-the-art prediction methods under the 5-fold cross-validation framework. The experimental results demonstrated that GTFKAN outperformed traditional methods in both AUC and AUPR metrics, highlighting its robust predictive capability. Moreover, we also explored the reason that why most models performed differently on these two well-known public databases of MDAD and aBiofilm. Finally, to further validate the model's practical effectiveness, we conducted

case studies on specific drugs and microorganisms extracted from the MDAD database, and results revealed that GTFKAN can accurately predict possible microbe-drug interactions and has significant application potential.

Data segmentation and cross validation

In this section, while adopting the 5-fold cross-validation framework to assess model performance, in order to ensure rational data partitioning and fairness in model evaluation, we would randomly disrupt the indexes of known and unknown associations respectively. And as a result, we would divide the previously downloaded association dataset into a training set and a test set with a ratio of 4:1. Moreover, when constructing the adjacency matrix and similarity matrix used for training, we would explicitly remove those known associations in the test set to ensure that the test data would not be used in the training process.

Hyperparameter sensitivity analysis

In this section, we would evaluate the effect of key parameters on performance of GTFKAN based on MDAD and aBiofilm databases by using the AUC and AUPR as indicators. Though observing above description of the GTFKAN, it is easy to see that GTFKAN has the following key parameters: the Dim-Hidden for the entity embedding dimension in Graph Transformers Networks, the edge features hidden-dim, the Num-Layers for the Transformer layers, the Num-Layers for the aggregated layers in the subtree K-hop, the Num-Heads for the polyhedral attention mechanism heads, and the Grid size of Fourier frequency in Fourier Kolmogorov-Arnold Networks.

For simplicity, we would first restrict the value of the entity embedding dimension Dim-Hidden in {64,128,256,512,1024}, the value of the Transformer's Num-Layers in {1,2,3,4}, the value of the sub-tree's aggregation layers (K-hop) in {1,2,3,4}, the value of the Edge features Hidden-dim in {64,128,256,512}, the value of the polyhedral attention mechanism's Num-Heads in {2,4,8,16}, and the value of the Fourier frequency grid size (Gridize) in {2,4,8,16}, respectively.

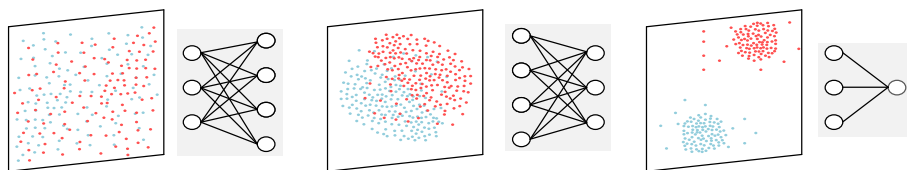


Figure 3. MLP Schematic. These representations become increasingly compact in each class, allowing for eventual linear separation.

And then, based on MDAD and aBiofilm databases, we assessed the effects of these parameters with different values on the performance of GTFKAN, and illustrated the AUC and AUPR values achieved by GTFKAN based on MDAD and aBiofilm databases in Figures 4 and 5 separately. From observing these figures, it is obvious that in the hyperparameter analysis, the median AUC and AUPR for GTN's entity embedding dimension increased consistently from 64 to 1024, achieving optimal performance and minimal fluctuation at 512 dimensions. The performance of GTN's neuron tree aggregation layers remained stable across 1–4 layers, with Layer 3 showing slightly better results. The number of Transformer layers demonstrated stable performance between 1 and 4, with Layers 1 and 2 showing marginally better outcomes. The performance of the Multihead Attention Mechanism improved between 2 and 16 heads, with the best results observed at 8 heads. Edge feature hidden dimensions maintained consistent performance between 64 and 512, with 256 dimensions identified as optimal. The Fourier frequency grid size in FKAN showed stable performance between 2 and 16, with size 8 yielding slightly better results.

Comparison with baseline methods

In this section, we further assess the predictive performance of GTFKAN by comparing it with five representative microbe-drug association prediction models:

- GLNNMDA, predicts microbe-drug interactions by integrating global and local features [25].
- MDASAE, combines stacked self-encoders and multi-head attention mechanisms for predicting microbe-drug associations [22].

- GCNMDA, based on graph attention networks and CNN classifiers, predicts interrelationships between microbes and drugs [53].
- GSAMDA, combines a sparse self-encoder with a graph convolutional attention network to identify potential microbe-drug associations [54].
- LCASPMMA, combines a learnable graph convolutional attention network with an adaptive iterative sampling strategy for predicting drug-drug relationships [24].

To ensure a fair comparison, we employed a 5-fold cross-validation (CV) framework with default parameters for all models, validating and analyzing them against each other in the MDAD and aBiofilm databases. The comparison results are presented in Table 2 below.

Figure 6 presents the optimal ROC and PR curves for these six competitive methods based on MDAD and aBiofilm data. Table 2 shows the AUC and AUPR values for these six competing methods based on MDAD and aBiofilm data. The figure and table indicate that GTFKAN outperforms the other five methods. The AUC values for GTFKAN are 0.9933 ± 0.0023 for MDAD and 0.9962 ± 0.0015 for aBiofilm, with AUPR values of 0.9487 ± 0.0024 and 0.9718 ± 0.0029 , respectively.

Furthermore, GLNNMDA ranks second to GTFKAN, exhibiting AUC values of approximately 98% across both datasets, yet its AUPR values are approximately 15% lower than those of GTFKAN. Despite GTFKAN's suboptimal AUPR on the aBiofilm dataset, it is just 0.02 below the optimal score achieved by LCASPMMA. GSAMDA exhibits the poorest performance among the deep learning models tested. In MDAD and aBiofilm, GSAMDA's AUC values are 0.9488 ± 0.0024 and 0.9353 ± 0.0034 , respectively, while the AUPR values are a mere 0.44520 ± 0.0013 and $0.4672 \pm$

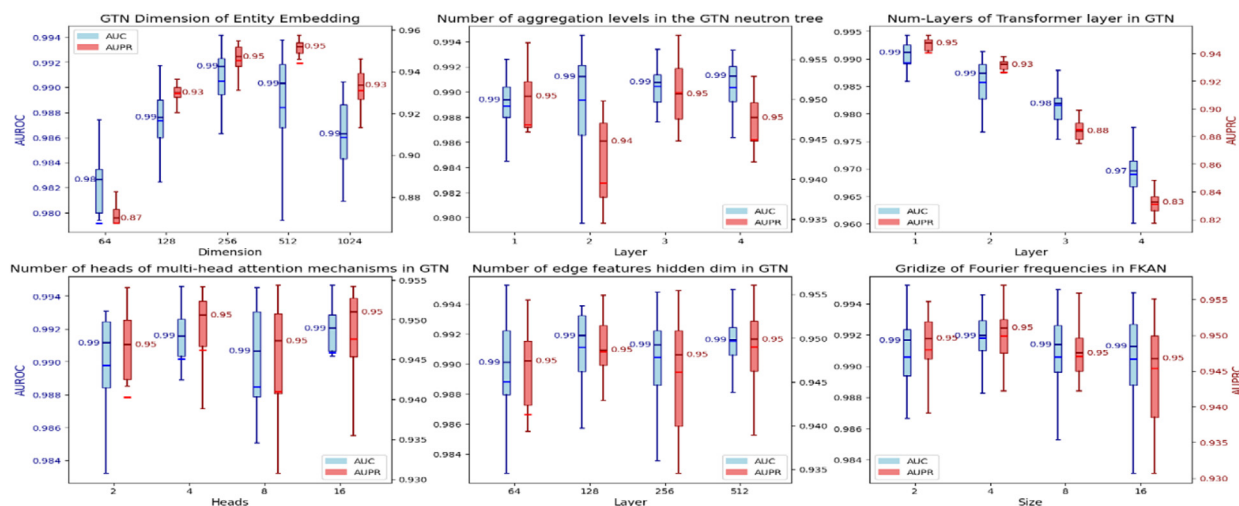


Figure 4. AUROC metrics and AUPRC metrics for each parameter of GTFKAN on the dataset MDAD.

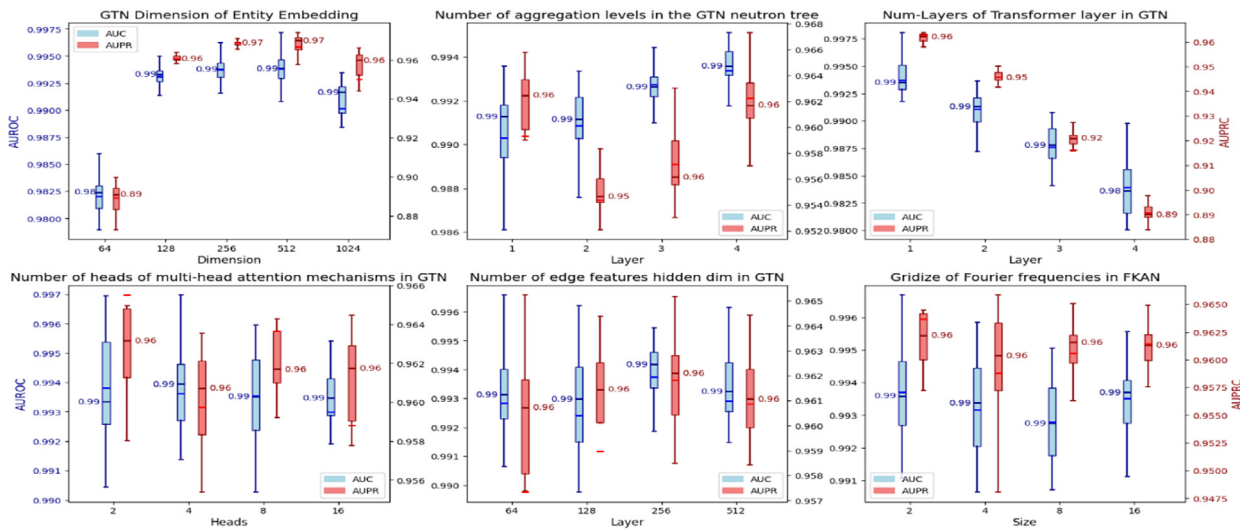


Figure 5. AUROC metrics and AUPRC metrics for GTFAKN parameters on the MDAD dataset.

Table 2 Results of the performance comparison between GTFAKN based on MDAD and aBiofilm databases and five competing methods at 5-fold CV.

Methods	AUC		AUPR	
	MDAD	aBiofilm	MDAD	aBiofilm
GLNNMDA	<u>0.9802 ± 0.0013</u>	<u>0.9773 ± 0.0023</u>	0.8586 ± 0.0015	0.8008 ± 0.0022
MDASAE	0.9565 ± 0.0023	0.9613 ± 0.0012	0.3655 ± 0.0024	0.3585 ± 0.0023
GSAMDA	0.9488 ± 0.0054	0.9353 ± 0.0014	0.4520 ± 0.0023	0.4672 ± 0.0041
LCASPMDA	0.9712 ± 0.0012	0.9763 ± 0.0037	0.9691 ± 0.0015	<u>0.9691 ± 0.0013</u>
GCNMDA	0.9549 ± 0.0042	0.9549 ± 0.0014	<u>0.9499 ± 0.0014</u>	0.9499 ± 0.0014
GTFAKN	0.9933 ± 0.0034	0.9961 ± 0.0034	0.9487 ± 0.0022	0.9718 ± 0.0016

Note: In this table, the best predicted values are shown in bold and the next best results are underlined.

0.0028, respectively. As depicted in Figure 6, while other models demonstrate good performance, GTFAKN remains the most competitive on these datasets.

To gain more insight into the distribution of predicted results, we extracted pre- and post-training learning representations from the trained GTFAKN model, which we projected into 2D space using t-SNE to visualise the training process. We show the results of the visualisation of the Roche test set in Figure 7, with microbes in red and drugs in blue. The comparison shows that the biological sample features are well extracted after performing the training.

Ablation study

In this section, we would further perform an ablation study on two variants of GTFAKN based on MDAD to explore the contribution of these modules to GTFAKN:

- GTFAKN without Graph Structure Feature Extractor (w/o GTN): in this variant of GTFAKN, we use only higher-order features of microbial-drugs learned from FKAN to make predictions.
- GTFAKN without Structural Feature Extractor (w/o FKAN): in this variant of GTFAKN, we apply only the microbe-drug heteromorphism maps learned in GTN as input to the predictor.

Table 3 illustrated the AUCs achieved by GTFAKN and its two variants based on MDAD. It is easy to see that the predictive power of GTFAKN would be affected when one of these two modules were removed, which means that all modules are essential for GTFAKN. As can be seen in Table 3, the GTFAKN (w/o FKAN) variant would be greatly affected when the structural features were removed, and its AUC value would be reduced by 8.34%. Meanwhile, the AUC value of GTFAKN (w/o GTN) would be decreased by 7.84%. Therefore, From the results of comparison

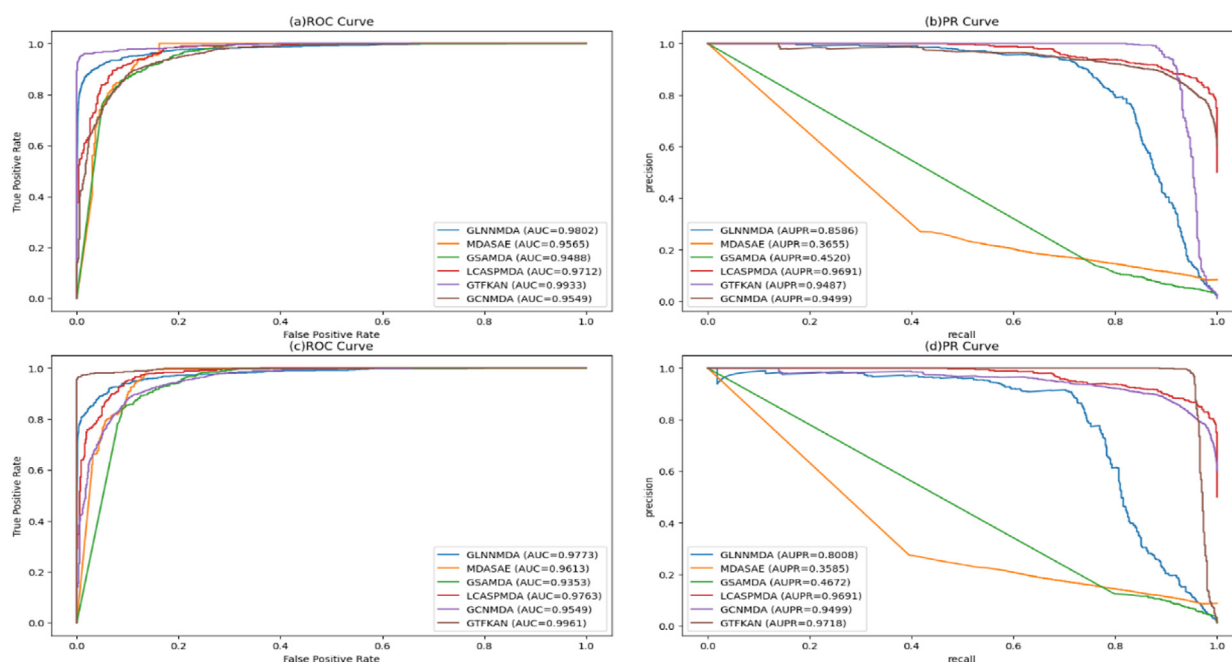


Figure 6. ROC and PR curves for GTFKAN and ROC and PR curves for the other five competing methods. (a) ROC curves of the six competing methods on MDAD. (b) PR curves of the six competing methods on MDAD. (c) ROC curves of the six competing methods on aBiofilm. (d) PR curves of the six competing methods on aBiofilm.

between GTFKAN (w/o FKAN) and GTFKAN (w/o GTN), it can be concluded that the application of structural feature information can significantly improve the predictive ability of GTFKAN. In addition, we also found that the removal of topological feature information would also lead to a decrease in model performance, which suggested that it was necessary to examine the topological information of the network.

Module analysis

In order to further verify the contribution of Graph Transformer Networks (GTN) to GTFKAN, in this section, we would select the Graph Attention Networks (GAT), which is also an attention mechanism, as the baseline for the control experiment in this study. In order to exclude the interference of structural features, the experiment

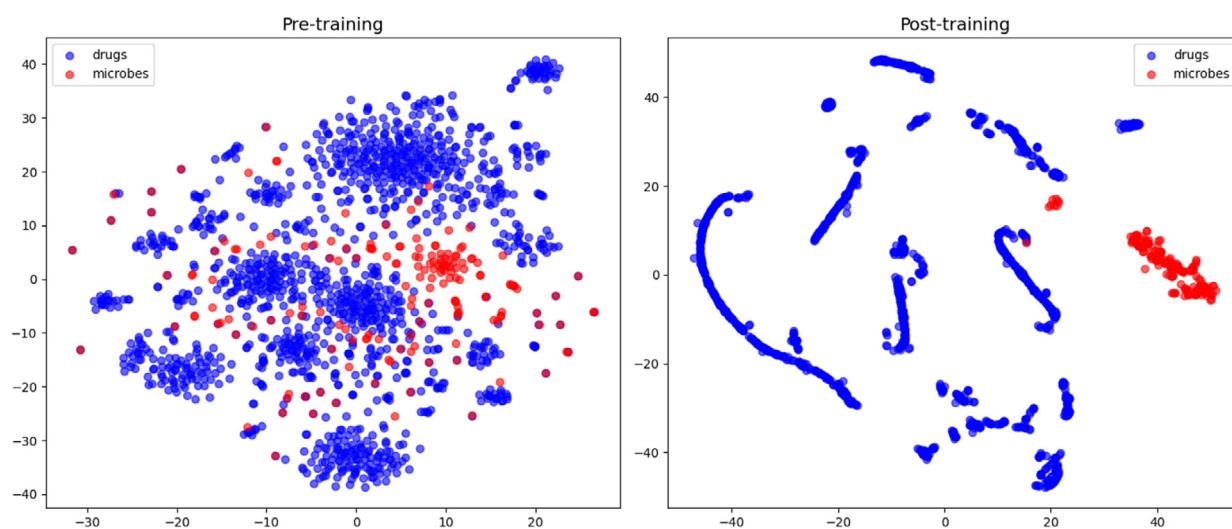


Figure 7. The t-SNE 2D visualisation of microbial and drug characterisation at different time periods in the proposed GTFKAN model.

Table 3 The AUC value for the ablation experiments are based on the data set MDAD comparison method at 5-fold CV.

Datasets	Set	AUC
MDAD	GTFKAN (w/o GTN)	0.9146
	GTFKAN (w/o FKAN)	0.9099
	GTFKAN	0.9933

only retains GTN and GAT as independent feature extractors for fair comparison under the same dataset division (consistent with the main experiment). The experimental results, as shown in Table 4, demonstrated that GTN had a significant advantage in the microbe-drug interaction prediction task: the mean AUC achieved by GTN-based GTFKAN reached 0.9017 (± 0.0095), which was 45.5% higher than that of 0.6199 (± 0.1701) achieved by GAT-based GTFKAN, and the standard deviation was reduced by 94.4%. Moreover, in terms of the AUPR metrics, GTN-based GTFKAN achieved (0.5089 ± 0.0271), which also outperformed the (0.2573 ± 0.0767) achieved by GAT-based GTFKAN by an absolute advantage of 97.8%, and the standard deviation was reduced by 64.67%. The detailed AUCs and AUPRs achieved by these two models were shown in Figure 8, based on which, it is easy to know that GTN has better generalization stability than GAT in GTFKAN.

The root of the performance advantage can be traced to the dynamic global sensing mechanism and multi-scale feature fusion capability of GTN. GTN is able to adaptively learn the dynamic topology of heterogeneous networks through the introduction of the Transformer's position encoding module, whereas GAT relies on a static aggregation strategy of fixed neighbors, which makes it difficult to capture non-local interaction patterns. GTN's multi-head self-attention mechanism (time complexity $O(N^2)$) allows each node to interact with the full graph nodes, breaking through the 1-hop localization restriction of GAT (time complexity $O(1)$). GTN automatically achieves multi-scale feature aggregation from local features to global feature interaction patterns by stacking multi-head attention layers. In contrast, GAT needs to manually design a multi-hop aggregation strategy and is prone to over-smoothing problems caused by stacking layers.

It is worth noting that the global awareness capability of GTN comes at the cost of

computational cost, and its training time increases by 522.82% compared with GAT at 10^7 node scale. This suggests that performance and efficiency need to be traded off in practical deployments, which can be optimized by sparse attention or chunked computation in the future.

Database analysis

To comprehensively validate the generalizability of the GTFKAN framework, this study was systematically evaluated on two independent drug-microbe association databases: the MDAD and aBiofilm. as shown in Table 2, GTFKAN demonstrated significant predictive performance, obtaining an AUPR value of 0.9718 on the aBiofilm database, while the MDAD database achieved an AUPR value of AUPR value of 0.9487, and the AUC values under both types of scenarios break through 0.99, which fully proves that the model has excellent differentiation ability.

To further explore the performance differences of GTFKAN in different data characteristic scenarios, we introduce multi-dimensional evaluation metrics for in-depth comparison (shown in Figure 9). The experimental results show that the performance advantage on the aBiofilm database is dominant. This difference may stem from the inherent characteristics of the data construction mechanism: aBiofilm is able to more accurately characterize biofilm-mediated resistance mechanisms by validating the captured microbial-drug interactions through biofilm formation experiments; whereas, MDAD, as a general-purpose database, suffers from an imbalance in the proportion of positive and negative samples, which may have a systematic impact on the prediction scores based on the.

To test the above hypotheses, we designed a knowledge-guided data enhancement strategy: the MDAD database was supplemented with positive samples based on domain knowledge by integrating the drug-drug interaction association network and the microbe-microbe interaction association network (shown in Figure 10). The enhanced validation experiments (shown in Figure 11) showed that the optimized data distributions resulted in different degrees of improvement in various index values, strongly confirming the potential impact of the original data characteristics on the model performance.

This provides a new paradigm shift in the field research: breaking away from the traditional performance enhancement path through

Table 4 Comparison of computational performance of GTN and GAT.

Model	AUC	AUPR	Complexity of time	An Epoch time(ms)
GTN	0.9017 ± 0.0095	0.5089 ± 0.0271	$O(N^2)$	7913.5
GAT	0.6199 ± 0.1701	0.2573 ± 0.0767	$O(1)$	1270.6

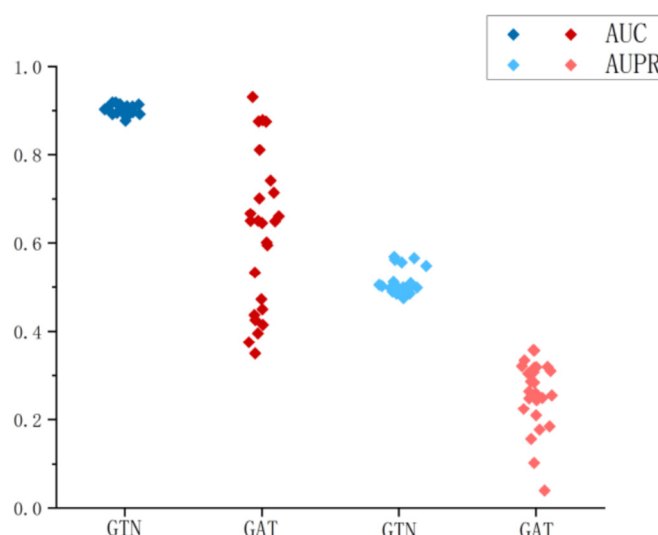


Figure 8. Comparative experimental data of GTN and GAT based on the dataset MDAD under 5-fold CV.

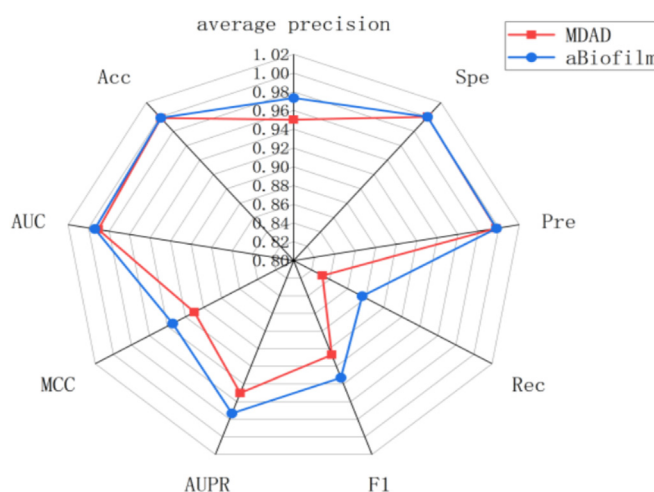


Figure 9. Comparative radar chart of GTFKAN's multidimensional assessment metrics on MDAD and aBiofilm databases.

algorithmic improvements only (e.g., similarity metric optimization, deep learning architecture tuning), and shifting to a data-algorithm co-evolutionary research framework. Specifically, the establishment of cross-scale correlations between microbial population sensing regulatory networks and drug-drug interaction networks can lead to improved bio-interpretability of predictive models.

Case study

To further validate the performance of GTFKAN in predicting unknown associations between microorganisms and drugs, in this section, we selected *Pseudomonas aeruginosa*, *Staphylococcus aureus*, and *Escherichia coli* as the case study subjects in this section. Specifically, we screened all candidate

microorganisms or drugs predicted to be associated with the selected targets according to GTFKAN. Finally, we screened and validated a randomised set of 10 microbes or drugs using published literature and investigated whether these associations were reported in the published literature. Since most of them behaved similarly in terms of name and predicted outcome. To avoid such results, 10 of these microbes or drugs were selected and screened based on the first letter of the name. One microorganism or drug was selected for each letter. The results of the case studies for *Pseudomonas aeruginosa*, *Staphylococcus aureus* and *Escherichia coli* are shown in [Tables 5–7](#).

Among the selected microorganisms and drugs, *Pseudomonas aeruginosa* is a Gram-negative bacterium that is widely found in water and soil

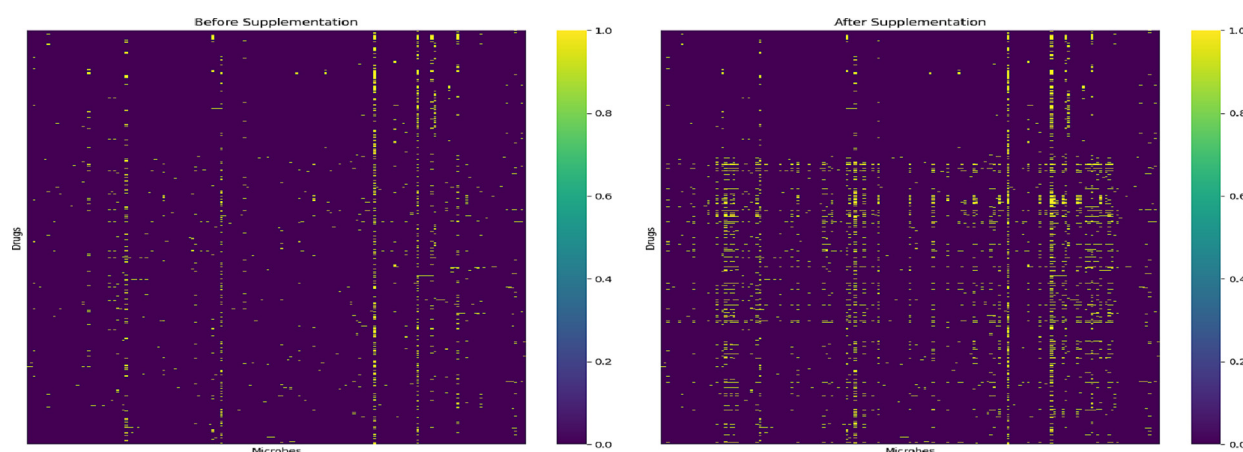


Figure 10. Heatmap of association of MDAD database before and after positive sample supplementation.

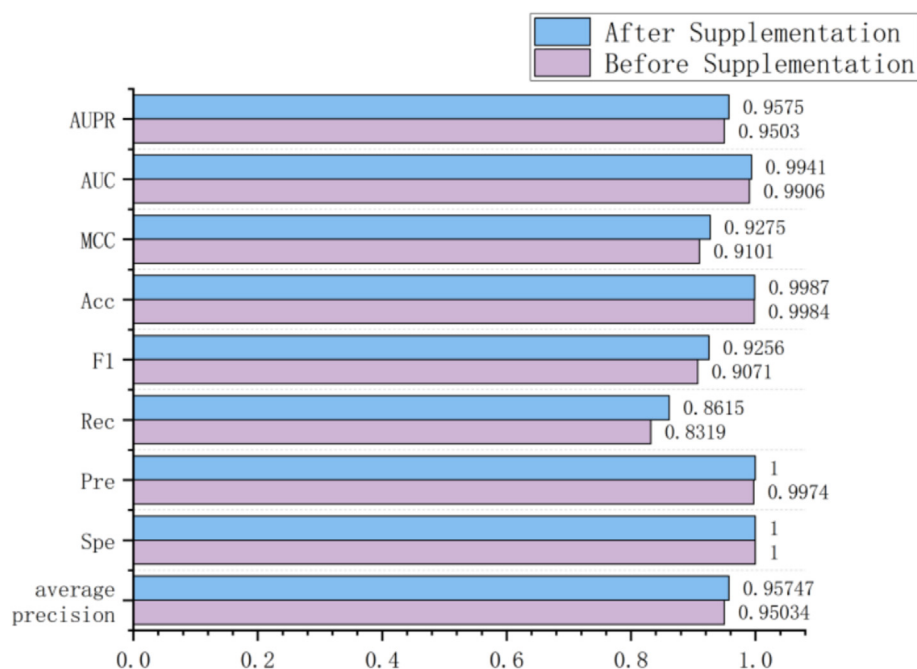


Figure 11. Indicators in the MDAD database before and after positive sample supplementation in the GTFKAN framework.

and has high metabolic adaptability and multiple virulence factors that can trigger infections in immunocompromised patients. It is naturally resistant to antibiotics and can be enhanced by genetic mutations. The resistance mechanisms of *P. aeruginosa* are complex and include the production of inactivating enzymes such as β -lactamases, alteration of drug targets (e.g., PBPs and DNA rotamases), reduction of outer membrane permeability, formation of biofilms, and reliance on an active efflux pumping system (dominant multidrug resistance). Against biofilm infections, macrolides (e.g., azithromycin) inhibit

biofilm formation and enhance the efficacy of other drugs. Previous studies have shown that *P. aeruginosa* is closely associated with many drugs [55]. Kandel et al. demonstrated [56] that *P. aeruginosa* contributes to the drug metabolism of ciprofloxacin, ivafloxacin and raloxifene.

Secondly, *Staphylococcus aureus* is a Gram-positive coccobacillus normally found in human skin and mucous membranes but can also cause a variety of infections ranging from minor skin infections to severe systemic infections. Its resistance, especially to methicillin-resistant *Staphylococcus aureus* (MRSA), makes treatment

Table 5 Ten drugs related to *Pseudomonas aeruginosa*.

\	drug	Evidence
1	Allicin	PMID: 31009577
2	Baicalin	PMID: 37058819
3	Caffeine	PMID: 31773197
4	Delphinidin	PMID: 27539815
5	Emodin	PMID: 37995454
6	Ferrous	PMID: 36321913
7	Gallium cation	PMID: 33705987
8	Heptadecanoic acid	PMID: 38633315
9	Indole	PMID: 36602321
10	Kojic acid	PMID: 39700872

Table 6 Ten drugs related to *Staphylococcus aureus*.

\	drug	Evidence
1	Aloe Vera Gel	PMID: 31391416
2	Baicalein	PMID: 37193303
3	Carvacrol	PMID: 28521511
4	Daptomycin	PMID: 29082727
5	Ellagic Acid	PMID: 36063129
6	Farnesol	PMID: 32224392
7	Gallic Acid	PMID: 37804166
8	Hesperetin	PMID: 39306988
9	Imipenem	PMID: 32921049
10	Kaempferol	PMID: 34856361

Table 7 Ten drugs related to *Escherichia coli*.

Indexing	Drug	Evidence
1	Amikacin	PMID: 35924928
2	Benzylpenicillin	PMID: 3060460
3	Caffeic Acid	PMID: 36621069
4	Doripenem	PMID: 36009912
5	Ertapenem	PMID: 28592240
6	Fosfomycin	PMID: 35024838
7	Gallic Acid	PMID: 36243382
8	Hesperidin	PMID: 33596837
9	Imipenem	PMID: 37255469
10	Kaempferol	PMID: 39723150

difficult. Numerous experiments have also confirmed the close relationship between *S. aureus* and drugs [57]. Ma et al. showed that [58], quercetin in combination with antibiotics acted as an inhibitor of *S. aureus* biofilm.

Finally, *Escherichia coli* is a common bacterium in the intestinal tract, most of which are harmless, but certain pathogenic strains can cause foodborne illnesses and other serious infections (e.g., extraintestinal infections and acute diarrhoea) and are resistant to certain drugs. Numerous experiments have also confirmed the close relationship between *Escherichia coli* and drugs. The synergistic effect of amikacin and rutin

enhances antibiotic penetration and accelerates bactericidal activity by disrupting *E. coli* cell membranes, while providing a new strategy for reducing antibiotic abuse and controlling drug resistance [59]. Teichmann et al. showed that [60], the evolutionary pathway of resistance to β -lactam antibiotics (e.g., amoxicillin) has been revealed through its gene mutation and stress response mechanism, providing a new opportunity for targeting the *dinB* or *katE* antimicrobial strategies, providing a theoretical basis.

Finally, we show in Figure 12 the network graph of the raw data and a network graph of the predicted results based on the 0.5 set threshold. The node size in the graph reflects the level of feature importance, and the node labels mark the key microorganisms involved in the case study. To further illustrate the degree of structural similarity between the two topological networks, the Structure Similarity Index Measure (SSIM) was chosen for comparison in this study, in which the SSIM value of these two network graphs was 0.8823. The high similarity indicates that the local structure (e.g., microbial-drug interaction patterns) is better preserved. The comparison shows that: the predicted network effectively preserved the core topological properties of the original network; the pivotal positions of key functional groups (e.g. *Pseudomonas aeruginosa* and *Escherichia coli*) in the network were not significantly shifted. The comparison validated the model's ability to resolve the topological features of the network, and showed that the prediction results could reliably reflect the biological significance of microbial interactions in the original data.

Discussion and conclusion

In this paper, we propose a novel computational model called GTFKAN by combining learnable Graph Transformers Networks, Fourier Kolmogorov-Arnold Networks, and multilayer perceptron. The biggest advantages of GTFKAN are in the following two aspects. Firstly, we utilise the GTN module for feature extraction on heterogeneous networks. Second, we use FKAN to extract the higher-order features of microorganisms and drugs in their lower-order feature matrices. The graph structure features and higher order features are then fused and final computation is performed using MLP. Comparative experiments and case studies show that GTFKAN can achieve excellent prediction performance. However, even though GTFKAN has excellent predictive ability, due to the imbalance of positive and negative samples in the dataset which has been missing deep information, we plan to go as far as possible to use the intricate interactions between microorganisms and drugs in our future work to improve the model performance. And try to develop a predictive

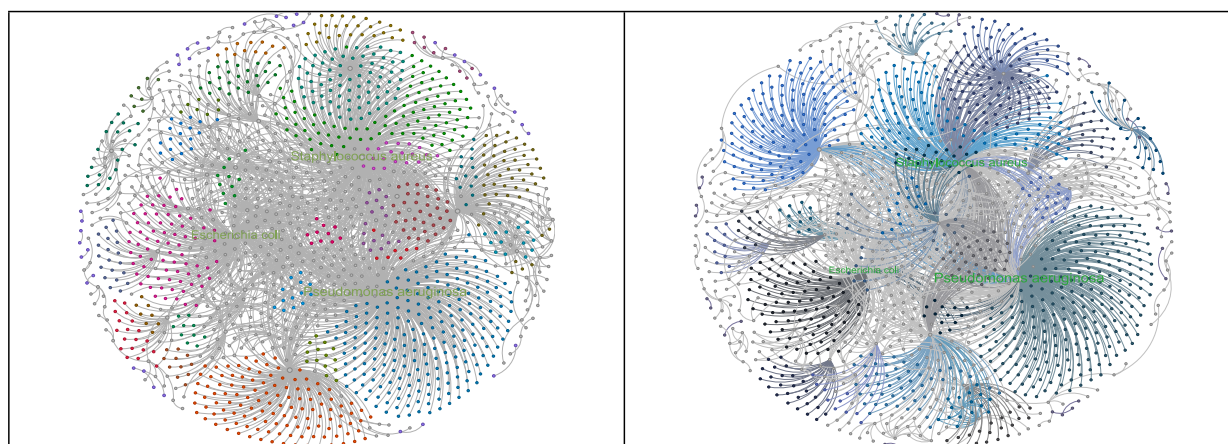


Figure 12. Topology of the original correlation network (left) vs. the predictive network (right).

model that can be applied to multiple domains (e.g., DDI, DTA, and other domains) rather than a single domain model.

Code availability

The source code and implementation details of GTFKAN are freely available at GitHub repository (<https://github.com/zjdssljc/GTFKAN.git>).

Data availability statement

The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding authors.

CRedit authorship contribution statement

Jiacheng Lai: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Zhen Zhang:** Project administration. **Bin Zeng:** Resources. **Lei Wang:** Writing – review & editing, Supervision, Project administration.

Funding

This work was partly sponsored by the National Natural Science Foundation of China (No. 62272064), the Natural Science Foundation of Hunan Province (No. 2023JJ60185) and the Key Project of Changsha Science and Technology Plan (No. KQ2203001).

DATA AVAILABILITY

Data will be made available on request.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the referees for suggestions that helped improve the paper substantially.

Received 26 March 2025;

Accepted 7 May 2025;

Available online 10 May 2025

Keywords:

microbe-drug association;
Graph Transformer Network;
Fourier Kolmogorov-Arnold Network;
the Gaussian kernel and functional similarities;
prediction model

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